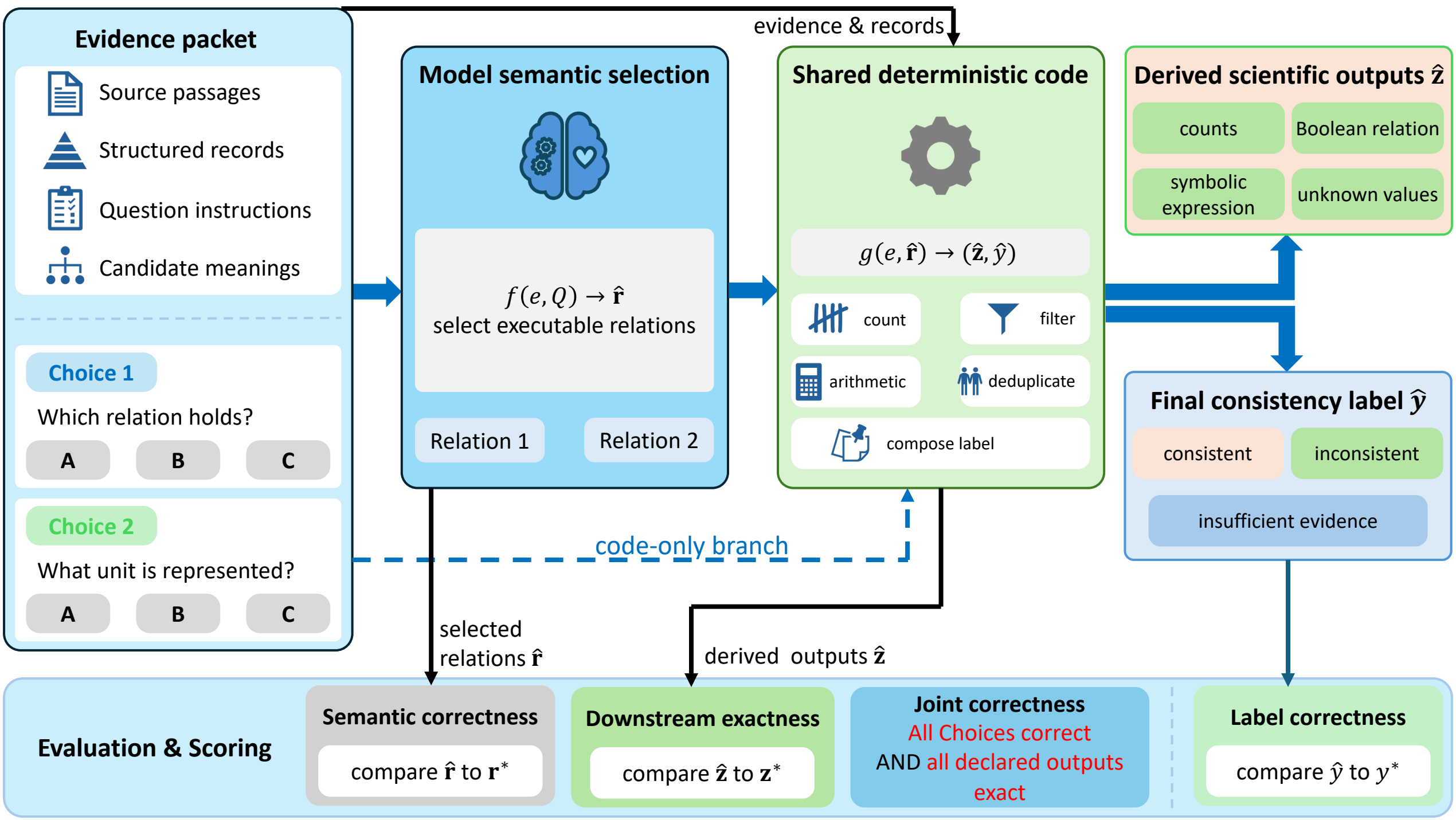

Evidence packet
Source passages
Structured records
Question instructions
Candidate meanings
Choice 1
Which relation holds?
A
B
C
Choice 2
What unit is represented?
A
B
C
evidence & records
Model semantic selection
$f(e, Q) \rightarrow \hat{\mathbf{r}}$
select executable relations
Relation 1
Relation 2
Shared deterministic code
$g(e, \hat{\mathbf{r}}) \rightarrow (\hat{\mathbf{z}}, \hat{y})$
count
filter
arithmetic
deduplicate
compose label
Derived scientific outputs $\hat{\mathbf{z}}$
counts
Boolean relation
symbolic expression
unknown values
Final consistency label $\hat{y}$
consistent
inconsistent
insufficient evidence
code-only branch
selected relations $\hat{\mathbf{r}}$
derived outputs $\hat{\mathbf{z}}$
Evaluation & Scoring
Semantic correctness
compare $\hat{\mathbf{r}}$ to $\mathbf{r}^*$
Downstream exactness
compare $\hat{\mathbf{z}}$ to $\mathbf{z}^*$
Joint correctness
All Choices correct
AND all declared outputs exact
Label correctness
compare $\hat{y}$ to y^*